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CVS Pharmacology

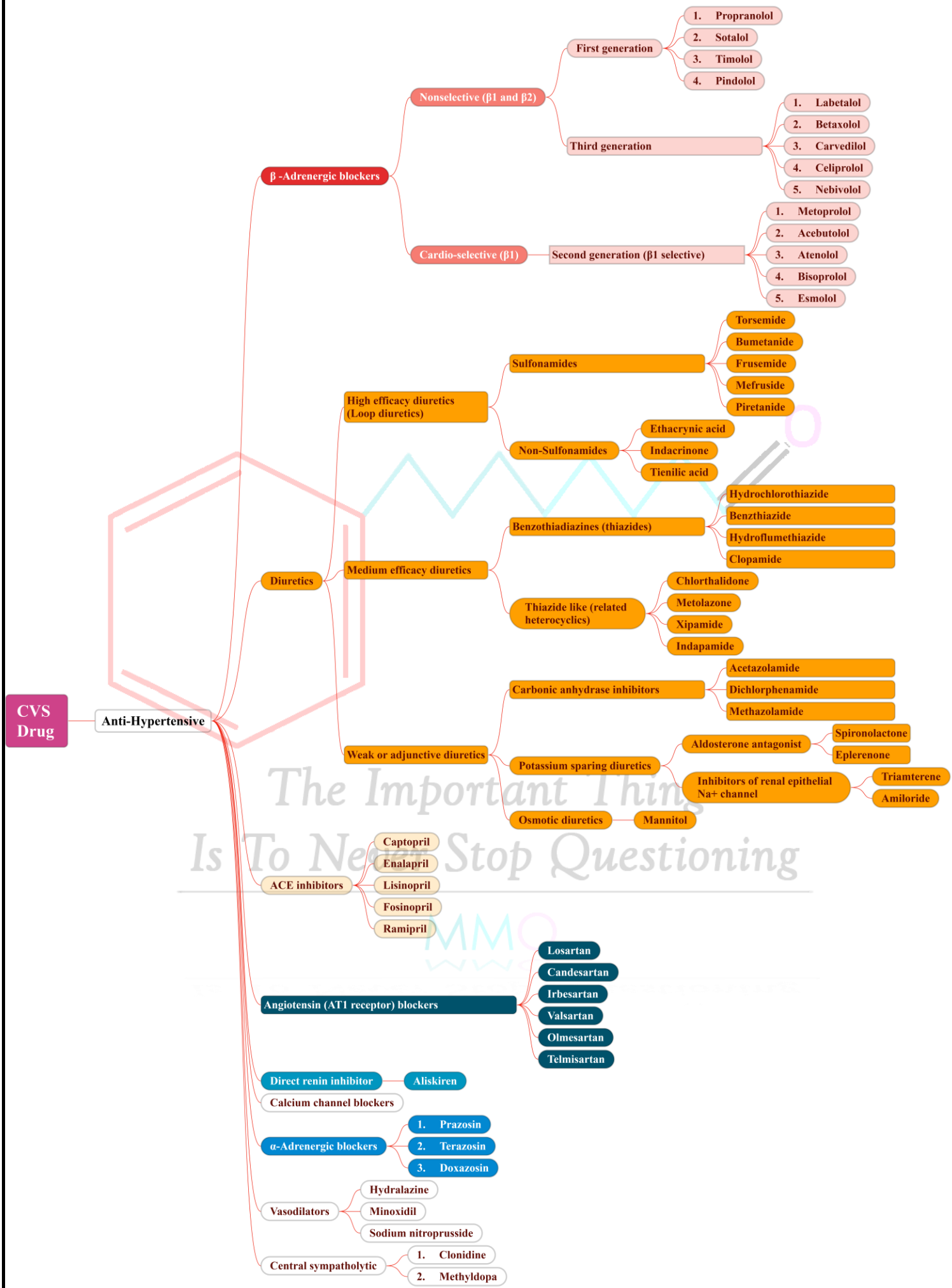


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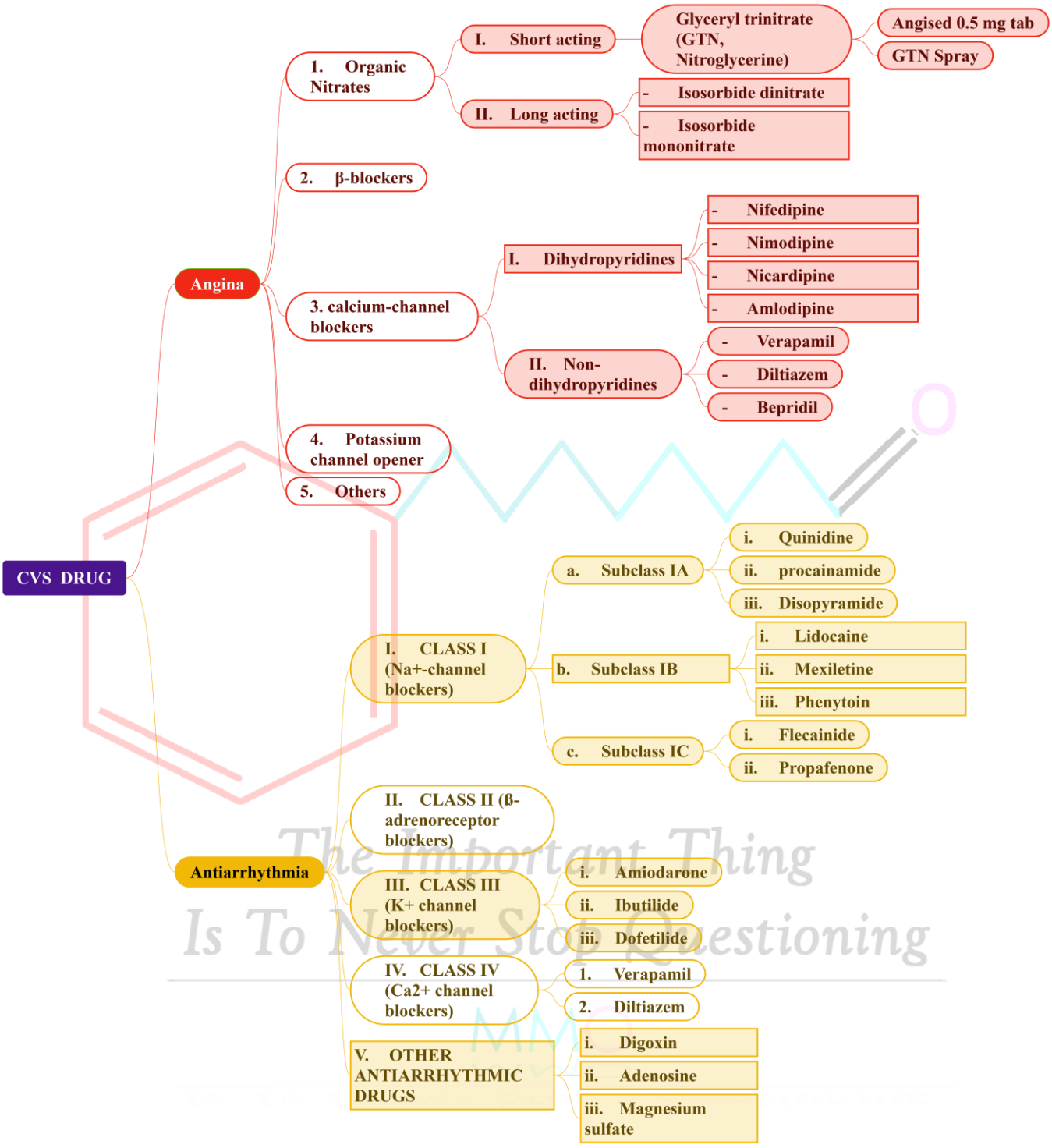
- الملف شامل للمقرر الدراسي مع الصور والمخططات التوضيحية المهمة
- سيوفر على الطالب عناء الحفظ، والبحث عن المعلومة، والوقت والجهد
- في الملف الإضافات المهمة باللغة العربية والإنجليزية من شرح الدكتور
- الملف مُنسق ومُراجع بصورة احترافية ومُنظمة بشكل رائع
- في الملف يوجد خرائط ذهنية تُسهل على الطالب حفظ الأدوية وعائلاتها بسهولة

Desined by:
#MMO_team

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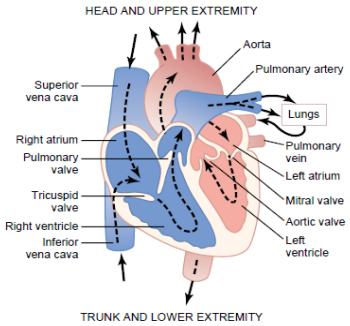


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Cardiovascular system



1. Heart
2. Blood vessels

1. Heart

- The heart is a muscular organ responsible for pumping blood through the circulation

• Anatomy of the heart

4 chambers

- 2 atria
- 2 ventricles
- Valves

1. The atria

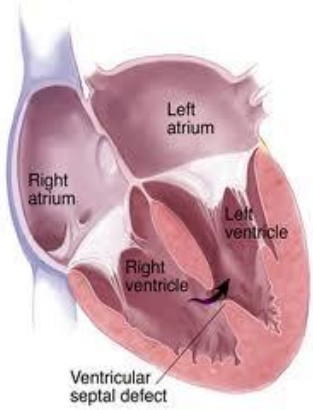
- Act mainly as a reservoir for blood and partially as a pump which Evacuates itself into the ventricles

2. The ventricles

- Act as a pump which ejects blood out of the heart into arteries

3. The Valves

1. 2 atrioventricular
   - Tricuspid
   - Mitral
2. 2 semilunar
   - Pulmonric
   - Aortic



• Electrical pace maker and conduction system

- S.A.node
- A.V. node
- His bundle and purkinje fibers

2. Blood vessels

I. Arteries

- Large diameters
- Thick-walled vessels that distribute the blood from the heart to the body

II. Arterioles

- Small thick wall vessels
- Represents major part of vascular resistance

III. Veins

- Large diameter (لذلك لا تحتاج لضغط أو ضخ لدفع الدم)
- Thin-walled vessels that carry the blood from the tissues to the heart

IV. Venules

- Small thin wall vessels
- Highly distensible

V. Capillaries

- Are small thin-walled vessels that allow exchange of gases and nutrients between the tissue cells and the blood.

3. Circulatory system

1. **Pulmonary circulation**
   - low pressure and resistance (the right heart pumps blood onto the pulmonary circulation)
2. **Systemic circulation**
   - High pressure and resistance (the left Heart pumps blood into the circulation)

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Hypertension

- **Hypertension**
  1. **Arterial blood pressure**
     - Is the pressure in the arterial wall measured in millimeters of mercury (mmHg).
  2. **Systolic pressure**
     - Systolic pressure the peak pressure exerted in the arteries when blood is pumped into them during ventricular systole.
  3. **Diastolic pressure**
     - Diastolic pressure the lowest pressure excreted in the arteries when blood is draining off into the vessels downstream during ventricular diastole.
- **Classification of hypertension**

BP Classification	Adult SBP (mm Hg)	Adult DBP (mm Hg)
Normal	Less than 120	and less than 80
Prehypertension	120–139	or 80–89
Stage 1 hypertension	140–159	or 90–99
Stage 2 hypertension	Greater than or equal to 160	Greater than or equal to 100

- **Types of hypertensions**
  1. Primary (essential) (unknown cause 90%)
  2. Secondary (due to known disease 10%)

Note:
Blood pressure = TPR X CO
TPR= total peripheral resistant
CO= cardiac output **"Where TPR increased:**
- Blood volume increased -Radius decreased

- **Causes of Secondary hypertension**
  1. **Diseases**
     - Thyroid diseases: أي خلل في الthyroid وخاصة زيادة Thyroxin يزيد من حساسية المستقبلات للأدرينالين وبالتالي يؤدي للـ Vasoconstriction، وزيادة بالـ CO & TPR وكل هذا يزيد من ارتفاع ضغط الدم.
     - Pheochromocytoma: سرطان في الغدة الكظرية adrenal medulla سيؤدي لنشاط في جهاز الSympathetic فيزداد إفراز الأدرينالين وبالتالي يؤدي للـ Vasoconstriction، وزيادة بالـ CO & TPR وكل هذا يزيد من ارتفاع ضغط الدم.
     - chronic kidney diseases: يزداد إفراز renin ويزداد معه إفراز angiotensin II فيزداد معه TPR وبالتالي يؤدي للـ Vasoconstriction، وكل هذا يزيد من ارتفاع ضغط الدم.
     - Primary aldosteronism: يزداد إفراز aldosterone ويؤدي ذلك لـ sodium-water retention فيزداد معه حجم الدم وبالتالي يؤدي لارتفاع ضغط الدم.
     - Cushing 's syndrome: Cortisone يزيد من عملية sodium-water retention فيزداد معه حجم الدم ونسبة الصوديوم، وبالتالي يؤدي لارتفاع ضغط الدم.
     - Coarctation of the aorta: هناك أمراض كثيرة مثل تصلب الشرايين، وأمراض القلب التي تؤدي لتضيق الشرايين بشكل عام....
  2. **Drugs**
     - Adrenal steroids e.g. prednisolone: تعمل على sodium-water retention فتزداد نسبة الصوديوم فيرتفع ضغط الدم.
     - Cocaine: يوقف عملية reuptake للأدرينالين سيؤدي لنشاط في جهاز الSympathetic فيؤدي للـ Vasoconstriction وزيادة الضغط.
     - Amphetamines: يعمل release للأدرينالين سيؤدي لنشاط في جهاز الSympathetic فيؤدي للـ Vasoconstriction وزيادة الضغط.
     - Contraceptive (estrogens): عبارة عن ستيرويدات تعمل على sodium-water retention فتزداد نسبة الصوديوم فيرتفع ضغط الدم.

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3. **Food substances**
   - Sodium يجب على المرضى تجنبه أو تقليل تناول الملح
   - Ethanol (alcohol)
   - Tyramine-containing foods if taking a MAO inhibitor: Tyramine يعتبر المادة الأساسية في صناعة الأدرينالين وهو حمض أميني يوجد في الزبدة مثلاً والأجبان وتسبب ارتفاع ضغط الدم المفاجئ (Hypertensive)

• Factors affecting blood pressure

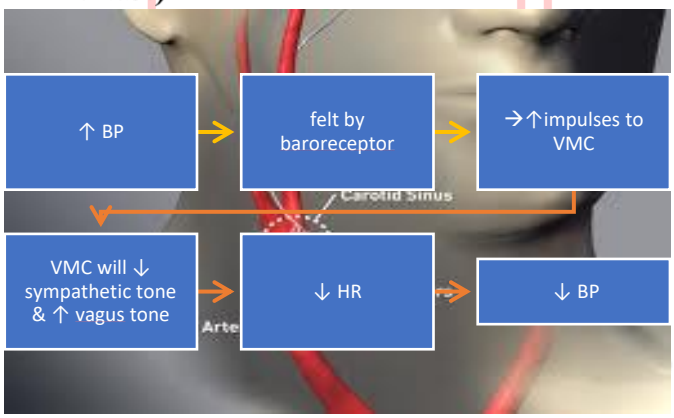
1. **Blood volume**
   - Increase blood volume = Increase blood pressure
   - Decrease blood volume = Decrease blood pressure
2. **Blood vessel radius**
   - Increase radius = Decrease blood pressure (V.D)
   - Decrease radius = Increase blood pressure (V.C)

The radius can be affected by physical activity, overall health, stress and hormones.

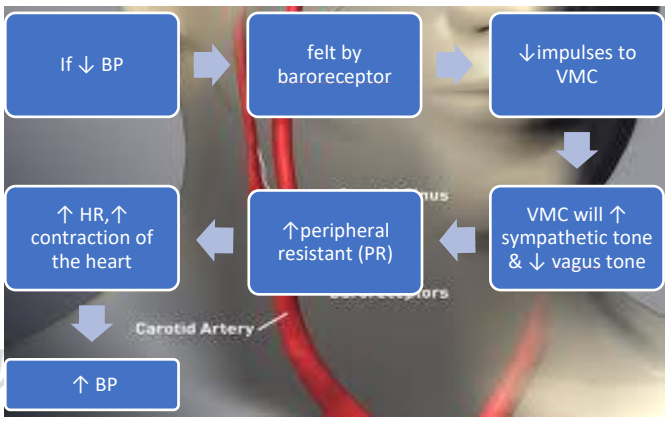
3. **Blood viscosity**
   - The thickness of blood due to decrease of water or anemia lead to increase of viscosity and blood pressure.
4. **Heart action**
   - Increase force = Increase blood pressure

• Regulation of blood pressure

i. Regulation of blood pressure short term regulation (seconds - to -minutes) (mediated by A.N.S.)



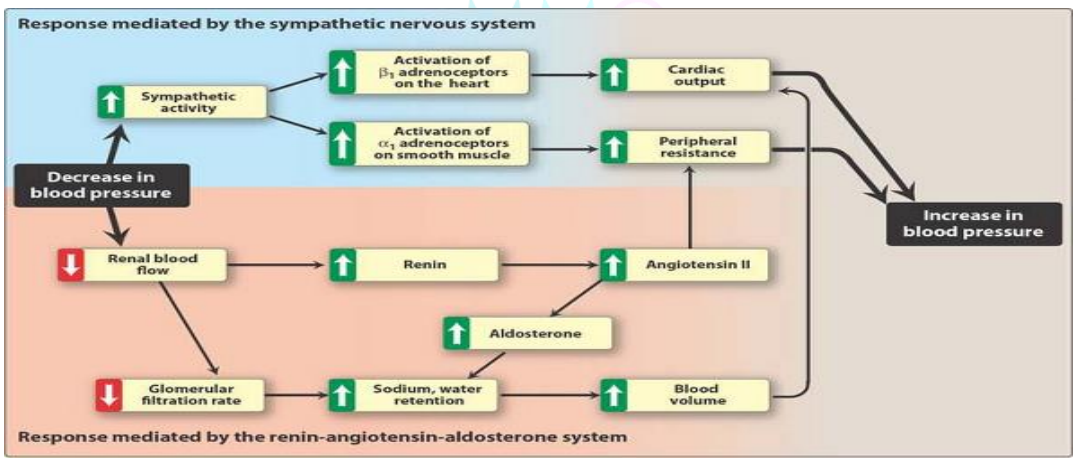
VMC- Vaso Motor Center



- Increase Vigus nerve = Bradycardia
- Decrease Vigus nerve= Tachycardia

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ii. Long term regulation (days to weeks)



- ## Regimen for Blood Pressure treatment

1. Lifestyle Modifications

- High sodium intake → Moderate sodium intake
- Excess consumption of calories → Weight reduction
- Physical inactivity → Regular aerobic activity
- Alcohol intake → Stop
- Smoking → Stop
- High stress→ Control stress

2. Antihypertensive drugs

i. Diuretics: من أجل تعمل تقليل لحجم الدم وتقليل كمية الدم للقلب والتي ستنقص الـ CO وسينخفض الضغط

ii. ACE inhibitors: تقلل من إفراز angiotensin II فيقل معه TPR وبالتالي يؤدي للـ Vasodilation، وكل هذا يقلل من ارتفاع ضغط الدم. أو يقل إفراز aldosterone فيقل معه TPRويؤدي ذلك لنقص حجم الدم وبالتالي يؤدي لانخفاض ضغط الدم.

iii. Angiotensin (AT1 receptor) blockers: كأنه يعمل blocking على مستقبلات α1

iv. Direct renin inhibitor: تقلل من إفراز renin فيقل angiotensin II فيقل معه TPR وبالتالي يؤدي للـ Vasodilation، وكل هذا يقلل من ارتفاع ضغط الدم.

v. Calcium channel blockers: تقلل Contractility of the heart فيؤدي إلى نقص CO وسينخفض الضغط، أما إذا اشتغل على الأوعية فإنه يعمل vasodilation ويقلل CO

vi. β -Adrenergic blockers: إما تعمل انخفاض CO أو تقليل إفراز الرنين أو انخفاض عمل جهاز الSympathetic فكل هذه تعمل على انخفاض ضغط الدم.

vii. β + α -Adrenergic blockers: تخفض CO أو TPR في القلب أو الأوعية فينخفض الضغط.

viii. α-Adrenergic blockers: يحدث Vasodilation فيقل TPR، وهذا يقلل من ارتفاع ضغط الدم.

ix. Central sympatholytic: تقلل من وظيفة القلب والأوعية وتنخفض CO أو TPR فينخفض الضغط.

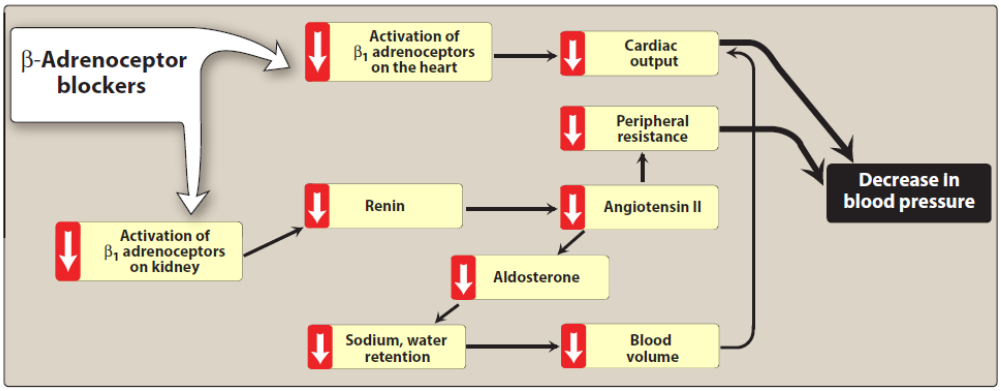
x. Vasodilators: تعمل على توسعة الأوعية فقط.

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i. β -adrenergic blocking drugs

- β-Blockers are a treatment option for hypertensive patient's concomitant heart disease or heart failure.

• Mechanism of action



• Classification

1. **Nonselective (β1 and β2)**

   i. First generation

   1. Propranolol
   2. Sotalol
   3. Timolol
   4. Pindolol

   ii. Third generation

   With additional α-blocking and/or vasodilator property

   1. Labetalol
   2. Betaxolol
   3. Carvedilol
   4. Celiprolol
   5. Nebivolol

2. **Cardio-selective (β1)**

   iii. Second generation (β1 – selective)

   1. Metoprolol
   2. Acebutolol
   3. Atenolol
   4. Bisoprolol
   5. Esmolol

• Pharmacokinetics

- The β-blockers are orally active
- Propranolol undergoes extensive and highly variable first-pass metabolism.
- Oral β-blockers may take several weeks to develop their full effects.
- Esmolol, metoprolol, and propranolol are available in intravenous formulations.

• Adverse effects

1. **Common effects**
   - Bradycardia
   - Hypotension
   - CNS side effects such as fatigue, lethargy, and insomnia
   - Decrease libido and cause erectile dysfunction
2. **Alterations in serum lipid patterns for** (non-cardio-selective β-blockers)
   - Decrease high-density lipoprotein cholesterol (HDL-C)
   - Increase Triglycerides (TG)

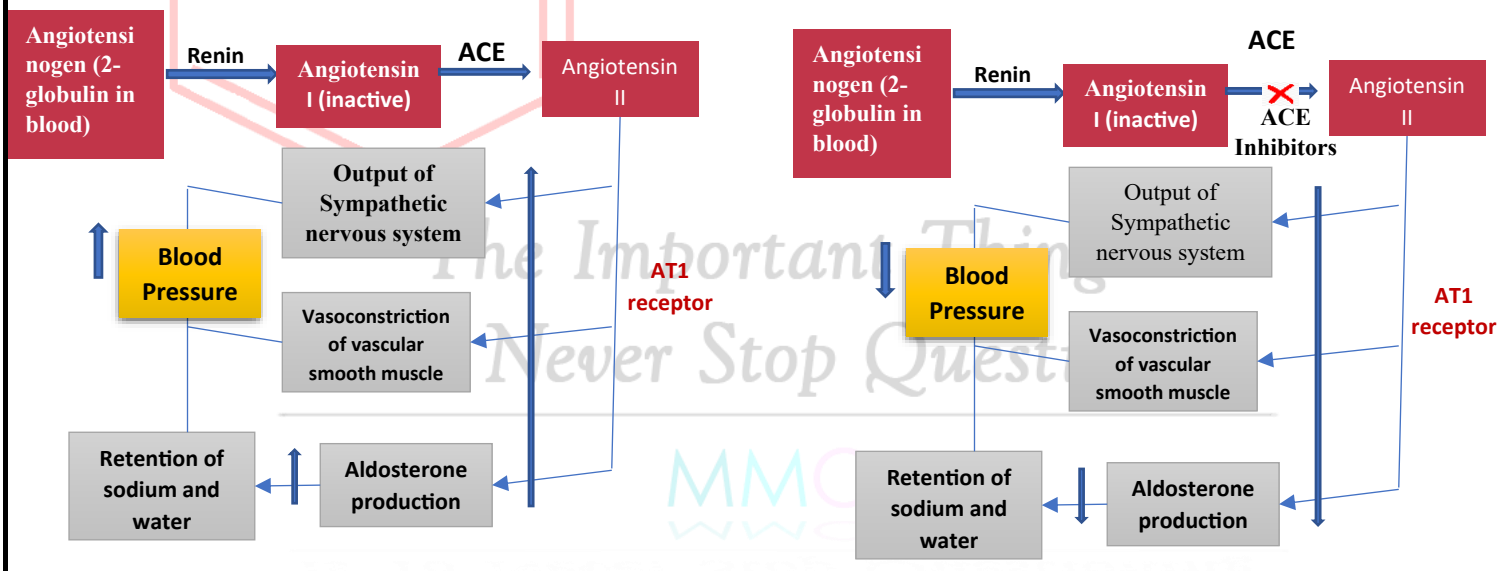
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3. **Drug withdrawal induce:**
   - Induce angina
   - Myocardial infarction (MI)
   - Sudden death in patients with ischemic heart disease.
4. **Block β2-Receptors** For **(Non-Selective β_2-Blockers):**
   - Bronchospasm & precipitate bronchial asthma (Treated by Theophylline)

• Therapeutic uses

1. **Hypertension**
   - β blockers are relatively mild antihypertensives.
2. **Angina pectoris**
   - Taken on a regular schedule they decrease frequency of attacks and increase exercise tolerance.
3. **Migraine**
   - Propranolol is the most effective drug for chronic prophylaxis of migraine
4. **Cardiac arrhythmias**
5. **Myocardial infarction (MI)**
6. **Congestive heart failure**
7. **Dissecting aortic aneurysm**
8. **Pheochromocytoma**
9. **Thyrotoxicosis** (↑ Tachycardia)
10. **Anxiety**
11. **Essential tremor**
12. **Glaucoma** (قافلات بيتا توقف عملية إنتاج السائل العيني)

ii. ACE Inhibitors



• ACE Inhibitors

Drug	Dose
Captopril	Daily dose (mg) 25–150
Enalapril	Daily dose (mg) 2.5–40
Lisinopril	Daily dose (mg) 5–40
Fosinopril	Daily dose (mg) 10–40
Ramipril	Daily dose (mg) 1.25–10

• Advantages of ACEI

1. Free of postural hypotension, electrolyte disturbances, feeling of weakness and CNS effects.
2. Safety in asthmatics, diabetics and peripheral vascular disease patients.
3. Long-term ACE inhibitor therapy has the potential to reduce incidence of type 2 diabetes in high-risk subjects.
4. Secondary hyperaldosteronism and K+ loss due to diuretics is prevented.
5. Renal blood flow is well maintained .
6. Left ventricular hypertrophy and increased wall-to-lumen ratio of blood vessels that occurs in hypertensive patients is reversed.
7. No hyperuricemia, no deleterious effect on plasma lipid profile.
8. No rebound hypertension on withdrawal.

rebound hypertension on withdrawal الذي تعملB-blockers بعكس أدوية

• Therapeutic uses

1. **Hypertension**
2. **Congestive heart failure (CHF)**
   - ACE inhibitors cause both arteriolar and veno-dilatation in CHF patients; reduce afterload as well as preload. بحيث يقلل الإجهاد وتوصيل كمية مناسبة وقليلة من الدم للقلب فيحسن من عملية ضغط الدم.
3. **Myocardial infarction MI**
4. **Prophylaxis in high cardiovascular risk subjects** كوقاية
5. **Diabetic nephropathy** كوقاية
   - Prolonged ACE inhibitor therapy has been found to prevent ordelay end-stage renal disease in type I as well as type II diabetics.

• Adverse effects

1. **Dry cough**
   - Due to accumulation of bradykinin in the lungs

Angiotensin II يسبب بتكسير bradykinin فعملنا تثبيط للـ Ag II وبالتالي لم يتكسر bradykinin وزادت كميته في الرئة وعمل bronchoconstriction و Dry cough.

2. **Rash**
3. **fever**
4. **altered taste**
5. **Hyperkalemia**
6. **ACE inhibitors are fetotoxic and should not be used by women who are pregnant.**

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iii. Angiotensin Antagonists

(Angiotensin receptor blockers or ARBs)

- These drugs block the AT1 receptors, decreasing the activation of AT1 receptors by angiotensin II.
- Their pharmacologic effects are similar to those of ACE inhibitors in that they produce arteriolar and venous dilation and block aldosterone secretion.
- They do not interfere with degradation of Bradykinin. Consequently, ACE inhibitor related cough is rare.

Drug	Dose
Losartan	Dose: 50 mg OD, rarely BD;
Candesartan	Dose: 8 mg OD (max 8 mg BD)
Irbesartan	Dose: 150–300 mg OD.
Valsartan	Dose: 80–160 mg OD 1 hour before meal
Olmesartan	Dose: 20–40 mg OD
Telmisartan	Dose: 20–80 mg OD

Note:

Addition of hydrochlorothiazide 12.5– 25mg enhances its effectiveness. (and balance of potassium level)

لأن hydrochlorothiazide يعمل Hypokalemia يعكس عمل الـ Hyper الذي يسببه ARBs فيحدث توازن لمستويات البوتاسيوم بالدم.

- **Therapeutic uses**
  - As ACE inhibitors
- **Adverse effects**
  - As ACE inhibitors except dry cough

iv. Direct Renin Inhibitor

- **Aliskiren**
  - Dose: 150–300 mg OD
  - It lowers blood pressure about as effectively as ARBs, ACE inhibitors, and thiazides.
  - Causes diarrhea, especially at higher doses, and can also cause cough and angioedema.
  - Contraindicated during pregnancy

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v. Central Sympatholytics

يثبط الجهاز السمبثاوي فيقل الضغط

1. **Clonidine**
2. **Methyldopa**

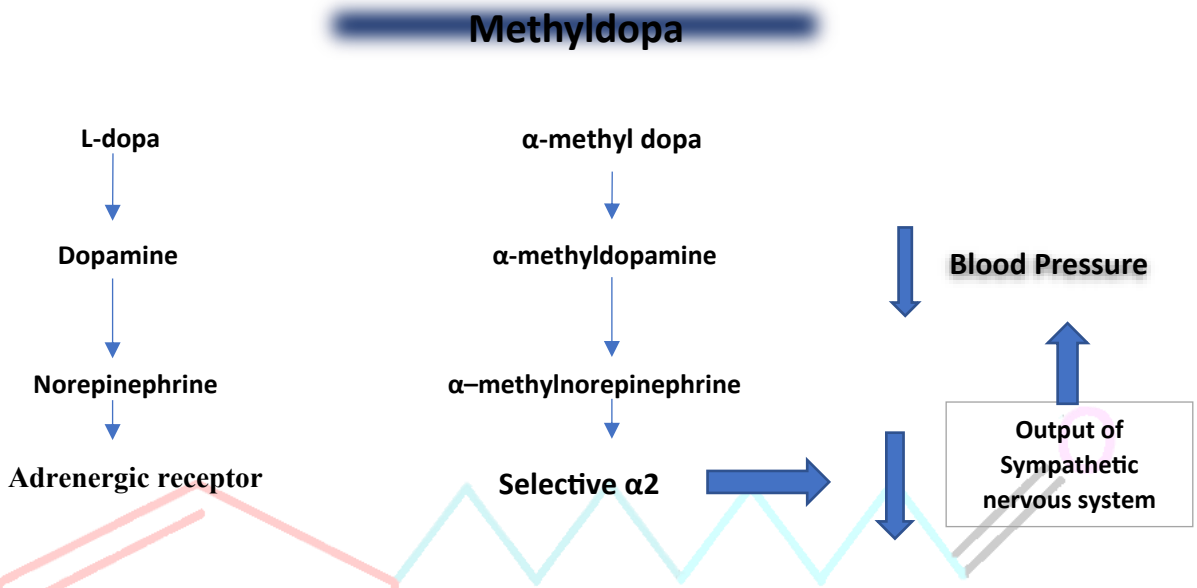
1. Clonidine

- Dose: Start with 100 µg OD or BD, max. 300 µg TDS, orally or i.m.
- Acts centrally as an α2 agonist → Produce inhibition of sympathetic vasomotor centers → ↓ sympathetic outflow to the periphery → ↓ Total peripheral resistance → ↓ blood pressure

- **Therapeutic uses of clonidine**
  - Clonidine is used primarily for the treatment of hypertension that has not responded adequately to treatment with two or more drugs.
- **Adverse effects**
  - Sedation, mental depression, disturbed sleep (CNS effect)
  - Dryness of mouth, nose and eyes
  - Rebound hypertension occurs following abrupt withdrawal

2. Methyldopa

- is an analog of L-dopa and is converted to α-methyldopamine and α–methylnorepinephrine
- Dose: 0.25–0.5 g BD–QID oral.
- It is mainly used for management of hypertension in pregnancy, where it has a record of safety.



vi. Vasodilators drugs

1. Hydralazine
2. Minoxidil
3. Sodium nitroprusside

1. Hydralazine

- Dose: 25–50 mg OD–TDS
  - It is a directly acting arteriolar vasodilator with little action on venous capacitance vessels.
  - reduces total peripheral resistance and causes greater decrease in diastolic than in systolic BP.

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Renin release → Increased aldosterone → Na+ and water retention → fluid retention and edema

Reflex compensatory mechanisms are evoked → Tachycardia

- Hydralazine is now used as a second line alternative only in combination with a diuretic and/or β blocker for patients not achieving target BP with first line drugs.
- It is almost always administered in combination with a β-blocker, such as propranolol (to balance the reflex tachycardia.
- It is almost always administered in combination with diuretics to overcome edema and fluid retention.

2. Minoxidil

- It is a powerful vasodilator, the pattern of action resembling hydralazine, i.e. direct relaxation of arteriolar smooth muscle with little effect on venous capacitance.
- Causes serious sodium and Water retention, Leading to volume overload, edema, and congestive heart failure.
- Causes hypertrichosis (the growth of body hair.)
- Is used in treatment of alopecia

3. Sodium nitroprusside

- It is a powerful parenterally administered vasodilator that is used in treating hypertensive emergencies as well as severe heart failure
- It is a rapidly (within seconds) and consistently acting vasodilator; has brief duration of action 2 - 5 min
- The action occurs as a result of release of nitric oxide.

vii. α- adrenergic blockers

1. Prazosin **2. Terazosin**

3. Doxazosin

1. Prazosin

- Selective α1 antagonist dilates both resistance and capacitance vessels.
- short-acting (3 times daily)
- unlike hydralazine, there is little reflex cardiac stimulation and renin release during long-term therapy.

• Side effects

- Severe postural hypotension
  Start by small dose (1mg) at bed time then increase the dose gradually.
- ↓ COP → ↓ RBF → ↓ Urine formation → Fluid retention (Add Diuretic)
- Failure of ejaculation

• Uses

- a moderately potent antihypertensive,
- Benign Prostatic Hyperplasia (BPH)

تعمل Blocking for α1A Receptors والذي يؤدي لـ Relaxation لطريق البول.

- Congestive heart failure

2. Terazosin, Doxazosin

- These are long-acting congeners of prazosin with similar properties but suitable for once daily dosing

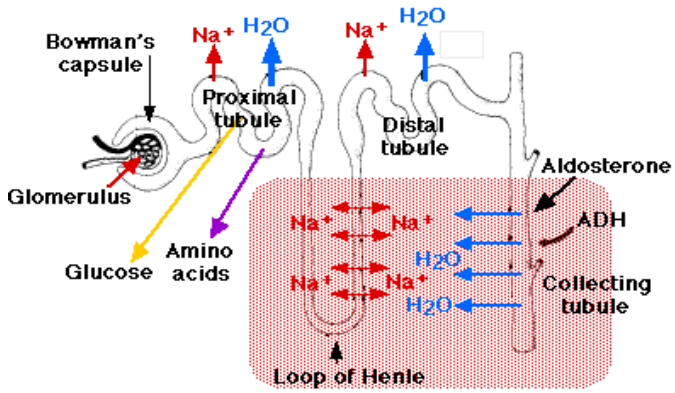
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viii. Diuretics

- Drugs that increase the volume of urine by increasing water & solute excretion.

This lead to reduce the blood volume, TPR, and blood pressure.

• Nephron



• Classification of Diuretics

1. **High efficacy diuretics**
2. **Medium efficacy diuretics**
3. **Weak or adjunctive diuretics**

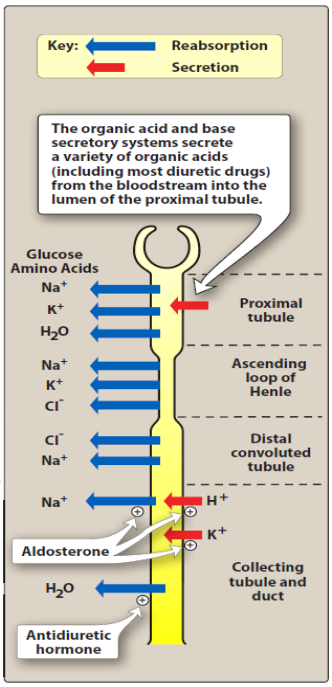
1. High efficacy diuretics (Loop Diuretics): according to the chemical structure:

I. Sulfonamides

1. Torsemide
2. Bumetanide
3. Frusemide
4. Mefruside
5. Piretanide

II. Non-Sulfonamides

1. Ethacrynic acid
2. Indacrinone
3. Tienilic acid



• Mechanism of action

- Loop diuretics inhibit Na+/k+/ Cl- co- transport system in the ascending loop of henle resulting in retention of Na+, Cl- and water in the tubule.

• Pharmacokinetics

- rapidly orally absorbed
- Hepatic metabolism
- They are eliminated by tubular secretion as well as by glomerular filtration

• Therapeutic uses

1. **Edema**
   The loop diuretics are the Drugs reducing of choice for The acute pulmonary edema of heart failure .
2. **Hypertension**
3. **Acute renal Failure** (i.e By the use of NSIADs- toxic substances)
4. **Hypercalcemia**
5. **Hyperkalemia**

عشان يعمل Hypokalemia فيحدث توازن لمستويات البوتاسيوم بالدم.

6. **Along with blood Transfusion in severe anaemia, To prevent vascular overload.**

To avoid hypervolemia

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• Drug Interactions

- Displaces Warfarin from plasma protein binding sites سيزداد تركيز Warfarin بالدم ويؤدي للنزيف
- Probenecid ↓ Renal tubular excretion of Frusemide
- N.S.A.I.D. ↓ Diuretic effect of Frusemide.
- Frusemide → ↑ Hypokalemia → ↑ Digitalis toxicity.
- Frusemide → Ototoxicity of Aminoglycosides e.g. Gentamicin.
- Frusemide → nephrotoxicity of Aminoglycosides & Cephalosporins.

• Adverse Effects & Toxicity

1. **Ototoxicity**
   Frusemide is toxic to hair cells of inner ear → Hearing defect & Deafness
2. **Hypokalemia**
   Worsens Digitalis Toxicity, and Liver & Kidney insufficiency.
3. **Hyperglycemia**
   Decrease Release of Insulin → Worsen Diabetes mellitus
4. **Hyperlipidemia**
   Increase Plasma Cholesterol & Triglycerides.
5. **Hyperuricemia**
   Decrease excretion of Uric acid → Worsens Gout.
6. **Hyponatremia**
7. **Hypochloremic Alkalosis.**
8. **Hypomagnesemia**
9. **Hypocalcemia**
10. **Hypovolemia, Hypotension & Dehydration**

❖ **Note:**

All the antihypertensive drugs cause hypotension as an adverse effect, Except (ACE inhibitors).

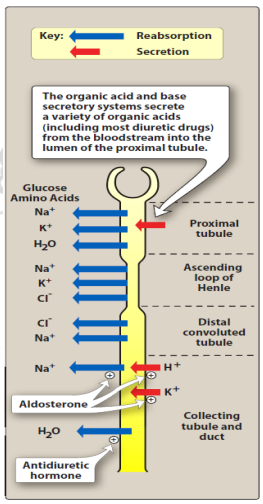
2. Medium efficacy diuretics

I. Benzothiadiazines (thiazides)

a. Hydrochlorothiazide
   (Used In combination with ARBs)
b. Benzthiazide
c. Hydroflumethiazide
d. Clopamide

II. Thiazide like (related heterocyclics)

a. Chlorthalidone
b. Metolazone
c. Xipamide
d. Indapamide

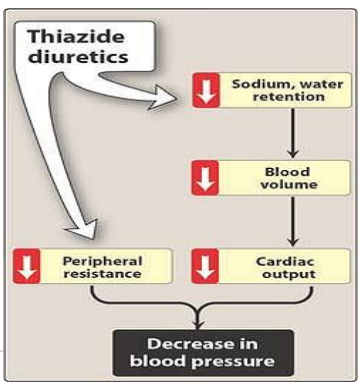


• Mechanism of action

- Thiazide diuretics ↑ the excretion of Na + and Cl– by inhibiting Na/Cl co transport. In the distal convoluted tubule i.e (↓ sodium and chloride reabsorption).
- Thiazide promote calcium reabsorption. →Hypercalcemia

• Pharmacokinetics

- All of the thiazides can be administered orally.
- All of the thiazides compete with the secretion of uric acid by that system. → Hyperuricemia
- As a result, uric acid secretion may be reduced, with an elevation in serum uric acid level.



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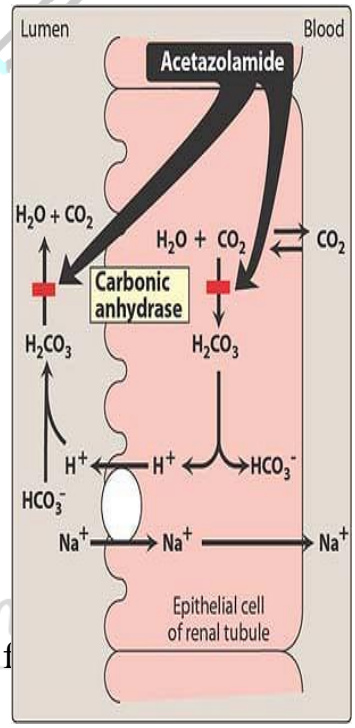
- **Therapeutic uses**
  1. **Mild to moderate Hypertension with normal renal function.**
  2. **Mild to moderate heart failure.**
  3. **Hypercalciurea**
  4. **Edema**
  5. **Diabetes insipidus** (Polyuria)
- **Side effects**
  1. **Hyperglycemia**
     Open K+-channel → Hyperpolarization → decrease Insulin release.
  2. **Hyperlipidemia**
     Increase Blood Cholesterol & Triglycerides.
  3. **Hypercalcemia**
  4. **Hypokalemia**
  5. **Hyperuricemia**
  6. **Hypomagnesemia**
  7. **Hyponatremia**
  8. **Hypovolemia**
  9. **Fetotoxic**

3. Weak or adjunctive diuretics

a. Carbonic anhydrase inhibitors
b. Potassium sparing diuretics
c. Osmotic diuretics

a. Carbonic anhydrase inhibitors

i. **Acetazolamide**
ii. Dichlorphenamide
iii. Methazolamide



1. Acetazolamide

- Inhibits carbonic anhydrase located intracellularly (cytoplasm) and on the apical membrane of the proximal tubular epithelium lead to decreased ability to exchange Na+ for H+ that H that give a mild diuresis.

- **Therapeutic Uses of Acetazolamide**
  1. As a diuretic BUT → Weak & Self-limiting.
  2. Glaucoma (inhibition of aqueous humor production)
  3. Petit Mal Absence epilepsy (absence seizures) when other drugs f
  4. Alkalinizer of urine e.g. to dissolve uric acid crystals.
- **Adverse effects**
  1. Metabolic acidosis (mild)
  2. Renal stone formation
  3. Potassium depletion (hypokalemia)

B. Potassium sparing diuretics

I. Aldosterone antagonist: Spironolactone
II. Inhibitors of renal epithelial Na+ channel :- Triamterene, Amiloride

I. Aldosterone antagonist:

1. Spironolactone (25,100 mg)

- It is a steroid, chemically related to the mineralocorticoid aldosterone

- **Mechanism of action of Spironolactone**
  - Antagonizes aldosterone at intracellular cytoplasmic receptor sites resulting in retention of K+ and excretion of Na+.

- **Therapeutic uses**
  - They are given with in combination with thiazide or loop diuretics in treating hypertension.
  - Prevent or correct hypokalemia
  - Used in patients taking digitalis to avoid potassium depletion
  - Hyperaldosteronism **(autoimmune or genetic disorders)**
  - Heart failure

- **Adverse effects**
  - Hyperkalemia
  - Gynecomastia لأن الستيرويدات غير اختيارية للمستقبلات وبالتالي قد ترتبط في مستقبلات الاستروجين
  - Impotence
  - menstrual irregularities

2. Eplerenone

- Is a recently developed more selective aldosterone antagonist
- less likely to cause hormonal disturbances like gynaecomastia, impotence and menstrual irregularities.

II. Inhibitors of renal epithelial Na+ channel

1. Triamterene
2. Amiloride

- **Nonsteroidal organic bases with identical actions.**

- **Therapeutic uses**
  - **Usually used in combination with Thiazide & loop diuretics to**
    a. Synergism
    b. Correct hypokalemia
    c. Correct hyperuricemia

- **Adverse effect**

1. Hyperkalemia
2. GIT disturbances
3. Allergy

C. Osmotic diuretics

- **Mannitol**

Intracranial pressure يستخدم في حالات

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Antianginal Drugs

كيف تحدث الـ Angina؟

يكون هناك نقص في تغذية الدم إلى القلب فتقل كمية الاكسجين الواصلة للقلب، فيزداد عمل القلب فيتحول الـ Respiration metabolism من هوائي إلى لا هوائي فتطلع مواد سامة مثل حمض اللاكتيك وبعض الـ Mediators هذه المواد تؤدي لظهور الالام شديدة وهي الاعراض الذي تظهر في الذبحة الصدرية.

• Angina pectoris

- Is a pain syndrome due to induction of an adverse oxygen supply/demand situation in a portion of the myocardium.
- is a characteristic by: sudden,severe,pressing chest pain radiating to the neck, jaw, back, and arms.
- **The imbalance between oxygen delivery and utilization may result-:**
  - During exertion
  - From a spasm of the vascular smooth muscle
  - From obstruction of blood vessels caused by atherosclerotic lesions.

• Types Of Anginas

- Classical angina (stable angina) (النوع الشائع)
- Variant/Prinzmetal/ Vasospastic angina (سببها يكون تضيق في الأوعية الدموية)
- Unstable angina (UA) (يدخل معها مرض MI)

• Aims of treatment

- Terminating or preventing an acute attack.
- Increasing the patient's exercise capacity
- These objectives can be achieved by:
  - Reducing overall myocardial oxygen demand
  - Increasing oxygen supply to ischemic areas.

Antianginal Drugs

The Important Thing Is To Never Stop Questioning

1. Organic Nitrates
2. β-blockers
3. calcium-channel blockers
4. Potassium channel opener
5. Others

1. Organic Nitrates

I. Short acting

- Glyceryl trinitrate (GTN, Nitroglycerine) تستخدم في حالة الـ acute وتأتي على شكل بخاخ أو تحت الفم

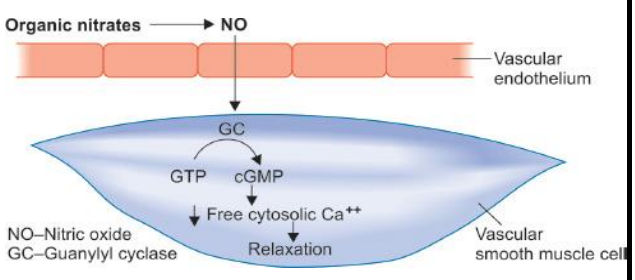
II. Long acting

- Isosorbide dinitrate
- Isosorbide mononitrate

تستخدم في حالة الـ prophylaxis (وقاية)

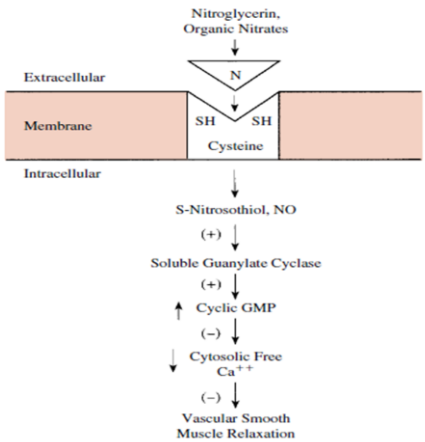
• Mechanism of Action of organic nitrates

- Decrease coronary vasoconstriction or spasm
  → Increase perfusion of the myocardium
- Dilate veins
  → Decreasing preload and myocardial oxygen consumption.



يحدث Releasing of NO فيقل الكالسيوم في العضلة وبالتالي يحدث Relaxation للعضلة

- **Pharmacokinetics**
  - Lipid soluble: well, absorbed from buccal mucosa, intestines and skin.
  - Ingested orally, all except isosorbide mononitrate (Buccal يستخدمو هن كـ)
  - Undergo extensive and variable first pass metabolism in liver.
- **Organic nitrats USES**
  - Angina pectoris
  - Acute coronary syndromes ACS
  - Cyanide poisoning يتفاعل NO مع السيانيد وينتج لنا Thiocyanate يكون أقل سمية وسهل للإخراج
  - Myocardial infarction MI
  - CHF and acute LVF
  - Biliary colic (Relaxation)
  - Esophageal spasm (Relaxation)
- **Organic nitrats Adverse effects**
  - Headache
  - postural hypotension
- **Organic nitrats Interactions**
  - Sildenafil causes dangerous potentiation of nitrate action: severe hypotension
  - MI and deaths are on record
- **Organic nitrats Tolerance**
  - May result from a reduced capacity of the vascular smooth muscle to convert nitroglycerin to NO due to depletion of of SH group



I. Glyceryl trinitrate (GTN, Nitroglycerine)

Preparations	Dose & route	Duration of action
Angised 0.5 mg tab	0.5 mg sublingual	10–30 min
GTN Spray	0.4–0.8 mg s.l. spray	10–30 min

- It acts within 1–2 min (peak blood level in 3–6 min) because of direct absorption into systemic circulation
- Nitroglycerine Is readily absorbed from the skin.
- A transdermal patch has been developed which provides steady delivery for 24 hours

II. Isosorbide dinitrate and Isosorbide mononitrate

Preparations	Dose & route	Duration of action
Isordil	5–10 mg sublingual	20–40 min
Isordil	10–20 mg oral	2–3 hr
Isosorbide dinitrate SR	20–40 mg oral	6–10 hr
Isosorbide mononitrate 10, 20, 40 mg tab	20–40 mg oral	6–10 hr

The Important Thing Is To Never Stop Questioning MMC

2. Calcium Channel Blockers (CCBs)

- **Calcium channels**
  1- Voltage sensitive channel (تفتتح وتتغلق حسب الشحنات الكهربائية)
  2- Receptor operated channel (مرتبطة بالـ receptors مثلاً تربط بمستقبلات النيكوتين)
  3- Leak channel (تعمل من أجل تعويض نقص الكالسيوم في القناة أو تخرج الكالسيوم الزائد من القناة)
- **Voltage sensitive channel**

- L-type	- T-type	- N-type
Present in contractile fibers يوجد فى جميع أنواع العضلات	Present in Nodal tissue and conducting system of the heart. يوجد في AV, AC node	Present in neuronal tissue يوجد في الأعصاب والدماغ

• Calcium Channel Blockers (CCBs) Uses

- Calcium channel blockers can be safely given to patients with obstructive lung disease and peripheral vascular disease in whom β blockers are contraindicated.
- Angina pectoris
- Benefit in classical angina appears to be primarily due to reduction in cardiac work
- Hypertension
- diltiazem and verapamil are among the first line drugs for hypertension.
- Cardiac arrhythmias
- Verapamil and diltiazem are highly effective in PSVT (Paroxysmal supraventricular tachycardia) and for control of ventricular rate in supraventricular arrhythmias

• Mechanism of action

- CCB blocks L-type Channels leading to inhibition of Ca+2 influx into smooth muscles leading to relaxation.

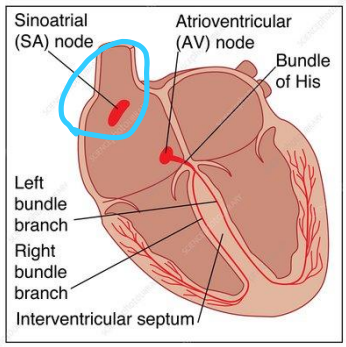
• classes of calcium channel blockers

I. Dihydropyridines	II. Non-dihydropyridines
- Nifedipine	- Verapamil
- Nimodipine	- Diltiazem
- Nicardipine	- Bepridil
- Amlodipine	

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I. Verapamil

- Block heart L-type
- Act mainly on heart
- ↓calcium entry into cardiac cells → ↓calcium intracellular & ↓ cardiac contractility and heart rate.
- Verapamil acts directly to depress S.A. node pacemaker activity and slow conduction. وبالتالي سيقل احتياج القلب للاكسجين
- Dose: 40–160 mg TDS oral, 5 mg by slow i.v. injection.



• Adverse effects

- Nausea
- Constipation لان الدواء يقلل motility of GIT
- Bradycardia

- **Interactions**
  - should not be given with β-blockers—additive sinus depression, conduction defects or asystole may occur **(Cardiac arrest سيسبب)**
  - It Increases plasma digoxin level by decreasing its excretion: toxicity can develop.
  - It should not be used along with other cardiac depressants like quinidine and disopyramide. **(Cardiac arrest سيسبب)**

II. Diltiazem

- Act On blood vessels & heart
- It is somewhat less potent vasodilator than nifedipine and verapamil.
- Usual clinical doses produce consistent fall in BP with little change or decrease in HR.
- Dose: 30–60 mg TDS–QID oral
- Interaction and Side effects are milder, but the profile is similar to verapamil.

III. Nifedipine

- Blocks L-type calcium Channels in vascular smooth muscles
- Act mainly on blood vessels (arteriolar dilatation) →Total peripheral resistance decreases → Blood pressure falls **لهذا الدواء يستخدم فقط في حالة HTN وليس في الـ Angina**
- It does not depress SA node or A-V conduction.
- Reflex sympathetic stimulation of heart predominates producing tachycardia, increased contractility.
- Dose: 5–20 mg BD–TDS oral.

- **Adverse effects**
  - Palpitation
  - Flushing
  - Ankle edema
  - Hypotension
  - Headache
  - Drowsiness and nausea

IV. Amlodipine

- It is the most distinct and the most popular.
- Oral absorption is slow, but complete; peak blood level occurs.
- Volume of distribution and t½ are exceptionally long: diurnal fluctuation in blood level is small and action extends over the next morning.
- Dose: 5–10 mg **هذا الدواء يستخدم صباح ومساء أو مرة فقط!**

V. Nimodipine

- It is a short-acting
- Penetrates blood-brain barrier very efficiently due to high lipid solubility.
- selectively relax cerebral vasculature and is approved for prevention and treatment of neurological deficit due to cerebral vasospasm following subarachnoid hemorrhage or ruptured congenital intracranial aneurysms. **هذا الدواء الوحيد الذي يعبرBBB ويعتبر فعال جداً.**

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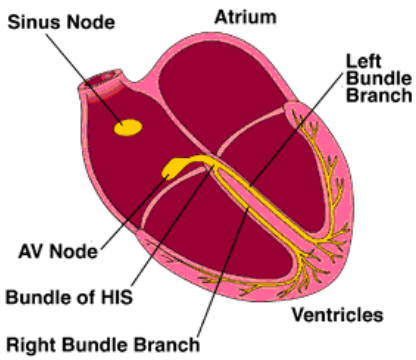
Antiarrhythmia

- **Arrhythmia**

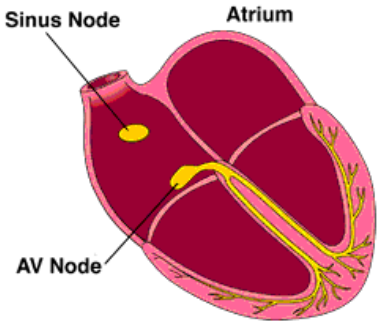
  Heart condition where disturbances in:
  - Pacemaker impulse formation
  - Contraction impulse conduction
  - Combination of the two

- **Normal heartbeat and atrial arrhythmia**

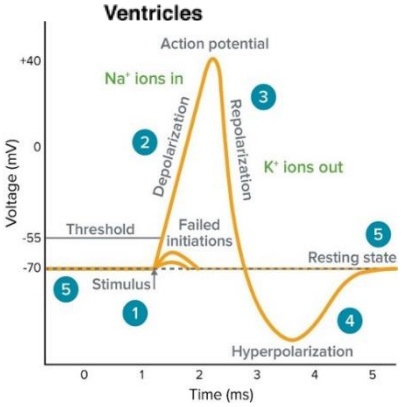
Normal rhythm

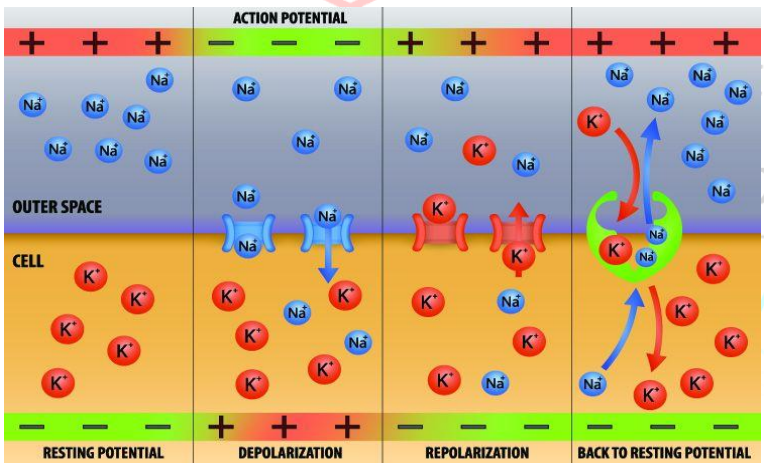


Atrial arrhythmia

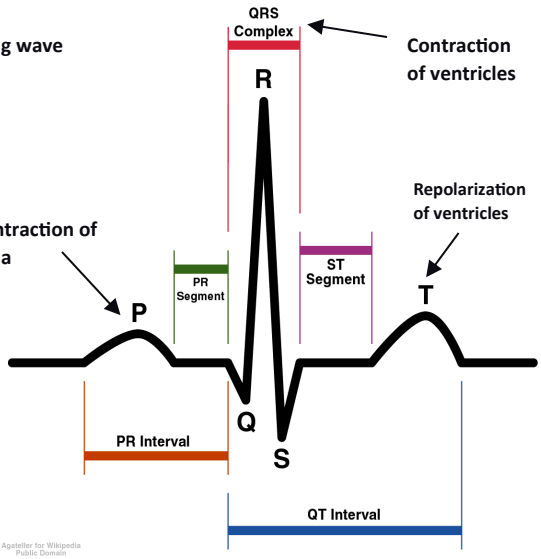


- **Electrophysiology - resting potential**
  - A transmembrane electrical gradient (potential) is maintained, with the interior of the cell negative with respect to outside the cell
  - Caused by unequal distribution of ions inside vs. outside cell
  - Na+ higher outside than inside cell
  - Ca+ much higher outside than inside cell
  - K+ higher inside cell than outside

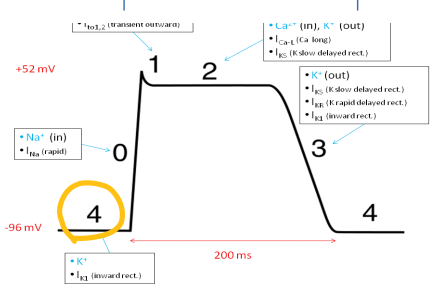




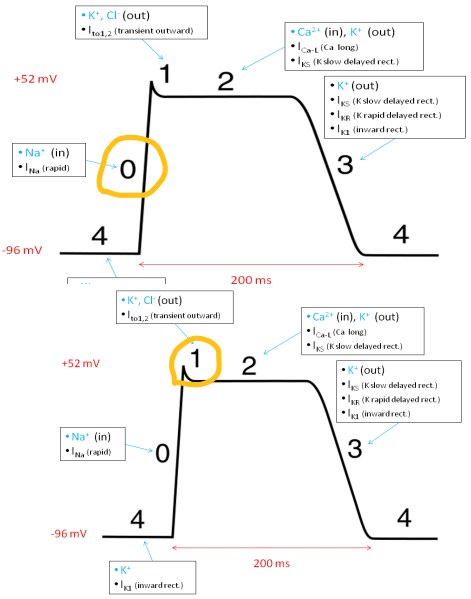
ECG (EKG) showing wave segments



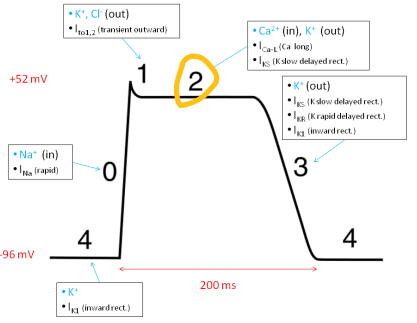
- **Cardiac Action Potential**
  - Divided into five phases (0,1,2,3,4)
    - **Cardiac Action Potential Phase 4 - resting phase (resting membrane potential)**
  - Phase cardiac cells remain in until stimulated
  - Associated with diastole portion of heart cycle



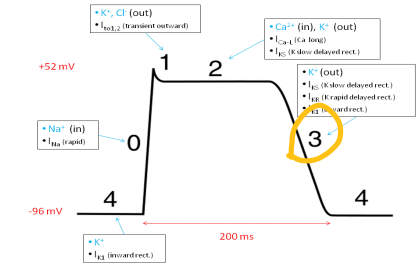
- **Addition of current into cardiac muscle (stimulation) causes**
  - Phase 0 – opening of fast Na channels and rapid depolarization
  - Drives Na+ into cell (inward current), changing membrane potential
  - Transient outward current due to movement of Cl- and K+



- **Cardiac Action Potential Phase 1 – initial rapid repolarization**
  - Closure of the fast Na+ Channels
  - Phase 0 and 1 together correspond to the R and S waves of the ECG



- **Cardiac Action Potential Phase 2 - plateau phase**
  - sustained by the balance between the inward movement of Ca+ and outward movement of K+
  - Corresponds to ST segment of the ECG



- **Cardiac Action Potential Phase 3 – repolarization**
  - K+ channels remain open, K+ outside the cell.
  - Corresponds to T wave on the ECG

- **Causes of arrhythmias**
  - Cardiac ischemia
  - Excessive discharge or sensitivity to autonomic transmitters
  - Exposure to toxic substances
  - Unknown etiology

The Important Thing
Is To Never Stop Questioning

Antiarrhythmic drugs

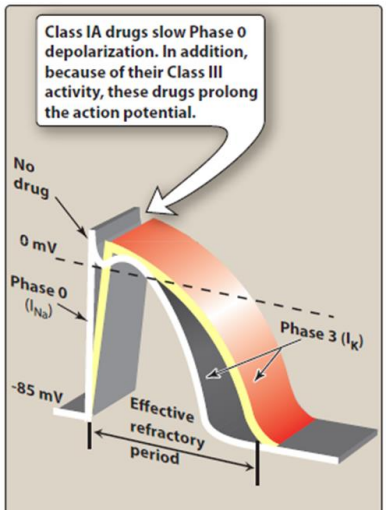
I. CLASS I (Na+-channel blockers)
II. CLASS II (ß-adrenoreceptor blockers)
III. CLASS III (K+ channel blockers)
IV. CLASS IV (Ca2+ channel blockers)
V. OTHER ANTIARRHYTHMIC DRUGS

I. Class I (Na+-channel blockers)

- Class I antiarrhythmic drugs act by blocking voltage- sensitive sodium (Na+) channels.
- **Include:**
  a. Subclass IA
  b. Subclass IB
  c. Subclass IC

a. Subclass IA

- Na+ channel blocker
- Cause moderate Phase 0 depression
- Include:
  - i.Quinidine
  - ii.procainamide
  - iii.Disopyramide
- Prolong repolarization
- Increased duration of action potential



i. Quinidine

- Quinidine binds to open and inactivated sodium channels and prevents sodium influx
- slowing the rapid upstroke during Phase 0
- Decreases the slope of Phase 4 spontaneous depolarization and inhibits potassium channels.

- **Pharmacokinetics**
  - Quinidine sulfate is rapidly and almost completely absorbed after oral administration.
  - It undergoes extensive metabolism by the hepatic cytochrome P450 enzymes, forming active metabolites.

- **Therapeutic uses**
  1. Atrial and ventricular
  2. arrhythmias

- **Side effects**
  1. torsades de pointes
  2. sudden cardiac arrest or VF
  3. idiosyncratic angioedema

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ii. Procainamide

- A derivative of the local anesthetic procaine
- shows actions similar to those of quinidine

- **Pharmacokinetics**
  - is well-absorbed following oral administration
  - The intravenous route is rarely used, because hypotension occurs if the drug is infused too rapidly

- **Side effects**
  - A reversible lupus erythematosus like syndrome that develops in 25 to 30 percent of patients.
  - Depression , hallucination, and psychosis.

iii. Disopyramide

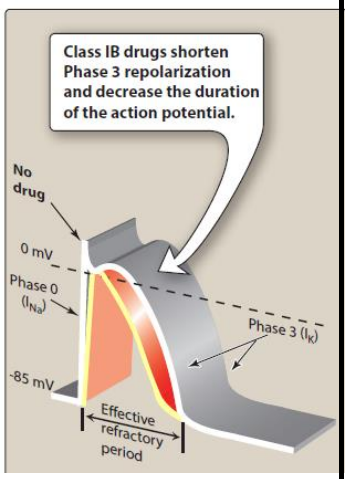
- It is a quinidine like Class IA drug that has prominent cardiac depressant and anticholinergic actions.

- **Pharmacokinetics**
  - Bioavailability of oral disopyramide is about .%80
  - It is partly metabolized in liver

- **Uses**
  - Treatment of ventricular arrhythmias as an alternative to procainamide or quinidine and may also be used for maintenance of sinus rhythm in atrial fibrillation or flutter.

b. Subclass IB

- These drugs block Na+ channels more in the inactivated than in the open state.
- Na+ channel blocker
- Weak Phase 0 depression
- include:
  - i. Lidocaine
  - ii. Mexiletine
  - iii. Phenytoin
- Shortened depolarization
- Decreased action potential duration



i. Lidocaine

- Lidocaine, is a local anesthetic
- blocks Na+ channels mostly in ventricular cells
- shortens Phase 3 repolarization and decreases the duration of the action potential
- also good for digitalis-associated arrhythmias
- Lidocaine is useful in treating ventricular arrhythmias arising during myocardial ischemia.
- is given only by i.v. route: 50–100 mg bolus followed by 20–40 mg every 20 –10 min or 1–3 mg/min infusion.

- **Adverse effects**

CNS effects include:

- Drowsiness
- slurred speech
- paresthesia
- agitation
- confusion, and convulsions.

ii. Mexiletine

- It is a local anesthetic and an orally active antiarrhythmic
- Chemically and pharmacologically similar to lidocaine.
- Mexiletine is Almost completely absorbed orally, 90% metabolized in liver and excreted in urine; plasma t½ 9–12 hours.
- Parenteral mexiletine may be used in post-infarction sinister ventricular arrhythmias as alternative to lidocaine.

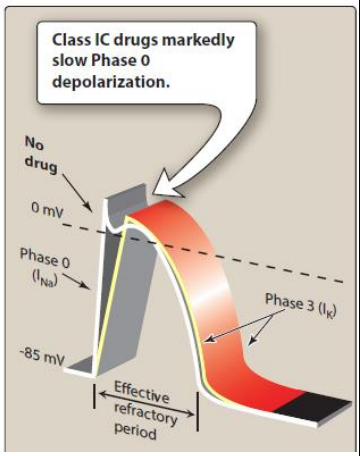
iii. Phenytoin

- Anticonvulsant That also works as antiarrhythmic similar to lidocaine

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MMO

c. Subclass IC

- These are the most potent Na+ channel blockers with more prominent action on open state
- Na+ channel blocker
- Strong Phase 0 depression
- include:
  - i. Flecainide
  - ii. Propafenone
- No effect of depolarization
- No effect on action potential duration



i. Flecainide

- Initially developed as a local anesthetic
- Flecainide is useful in the maintenance of sinus rhythm in atrial flutter or fibrillation in patients without structural heart disease
- Flecainide has a negative inotropic effect and can aggravate chronic heart failure

ii. Propafenone

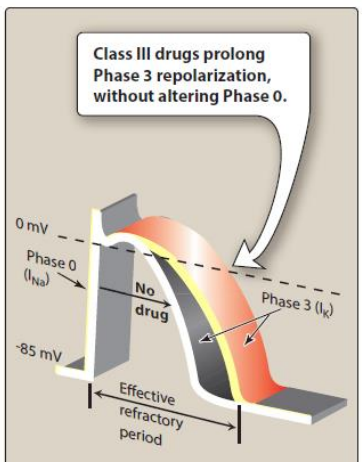
- Propafenone is used for prophylaxis and treatment of ventricular arrhythmias.
- Propafenone is absorbed orally and undergoes variable first pass metabolism

- **Side effects**
  1. nausea,
  2. vomiting, bitter taste,
  3. constipation and blurred vision.

II. Class II (ß-adrenoreceptor blockers)

- Causes both myocardial β– adrenergic blockade and membrane-stabilizing effects
- Slows SA node and ectopic pacemaking
- Can block arrhythmias induced by exercise or Apprehension.

III. Class III (K+ channel blockers)



- K+ channel blockers
- diminish the outward potassium current during repolarization of cardiac cells
- prolong the duration of the action potential without altering phase 0 of depolarization or the resting membrane potential
- **include:**
  - i. Amiodarone
  - ii. Ibutilide: Slows Inward movement of Na+ in addition to delaying K +influx
  - iii. Dofetilide: Prolongs action potential by delaying K+ efflux with no other effects

i. Amiodarone

- Contains iodine and is related structurally to thyroxine.
- It has complex effects, showing class I, II, III, and IV actions, as well as á-blocking activity.
- Its dominant effect is prolongation of the action potential duration and the refractory period by blocking K+ channels.

- **Therapeutic uses**
  - Treatment of severe refractory supraventricular and ventricular tachyarrhythmias.
  - Amiodarone has been a mainstay of therapy for the rhythm management of atrial fibrillation or flutter.

- **Pharmacokinetics**
  - Amiodarone is incompletely absorbed after oral administration.
  - having a prolonged half-life of several weeks, and it distributes extensively in adipose tissue.
  - Full clinical effects may not be achieved until months after initiation of treatment

- **Adverse effects**
  - Pulmonary fibrosis
  - Neuropathy
  - Hepatotoxicity
  - blue-gray skin discoloration, and hypo- or hyperthyroidism.
  - Amiodarone
  - Optic neuritis

IV. Class IV (Ca2+ channel blockers)

1. **Verapamil**
   - blocks Na+ channels in addition to Ca2+; also slows SA node in tachycardia
2. **Diltiazem**

V. Other Antiarrhythmic Drugs

i. Digoxin
ii. Adenosine
iii. Magnesium sulfate

i. Adenosine

- Decreases conduction velocity
- Prolongs the refractory Period
- Decreases automaticity in the AV node.
- Intravenous adenosine is the drug of choice for abolishing acute supraventricular tachycardia
- Adenosine has an extremely short duration of action (approximately 10 to 15 seconds) due to rapid uptake by erythrocytes and endothelial cells.

ii. Magnesium sulfate

- Magnesium is necessary for the transport of sodium, calcium, and potassium across cell membranes.
- It slows the rate of SA node impulse formation and prolongs conduction time along the myocardial tissue.
- Intravenous magnesium sulfate is the salt used treat arrhythmias,
- Most notably, magnesium is the drug of choice for treating the potentially fatal arrhythmia torsades de pointes and digoxin-induced arrhythmias.

Pacemakers

- Surgical implantation of electrical leads attached to a pulse generator

The Important Thing Is To Never Stop Questioning

Congestive Heart Failur (CHF)

➢ CHF – the heart is unable to pump sufficient blood to meeting the needs of the body

Symptoms

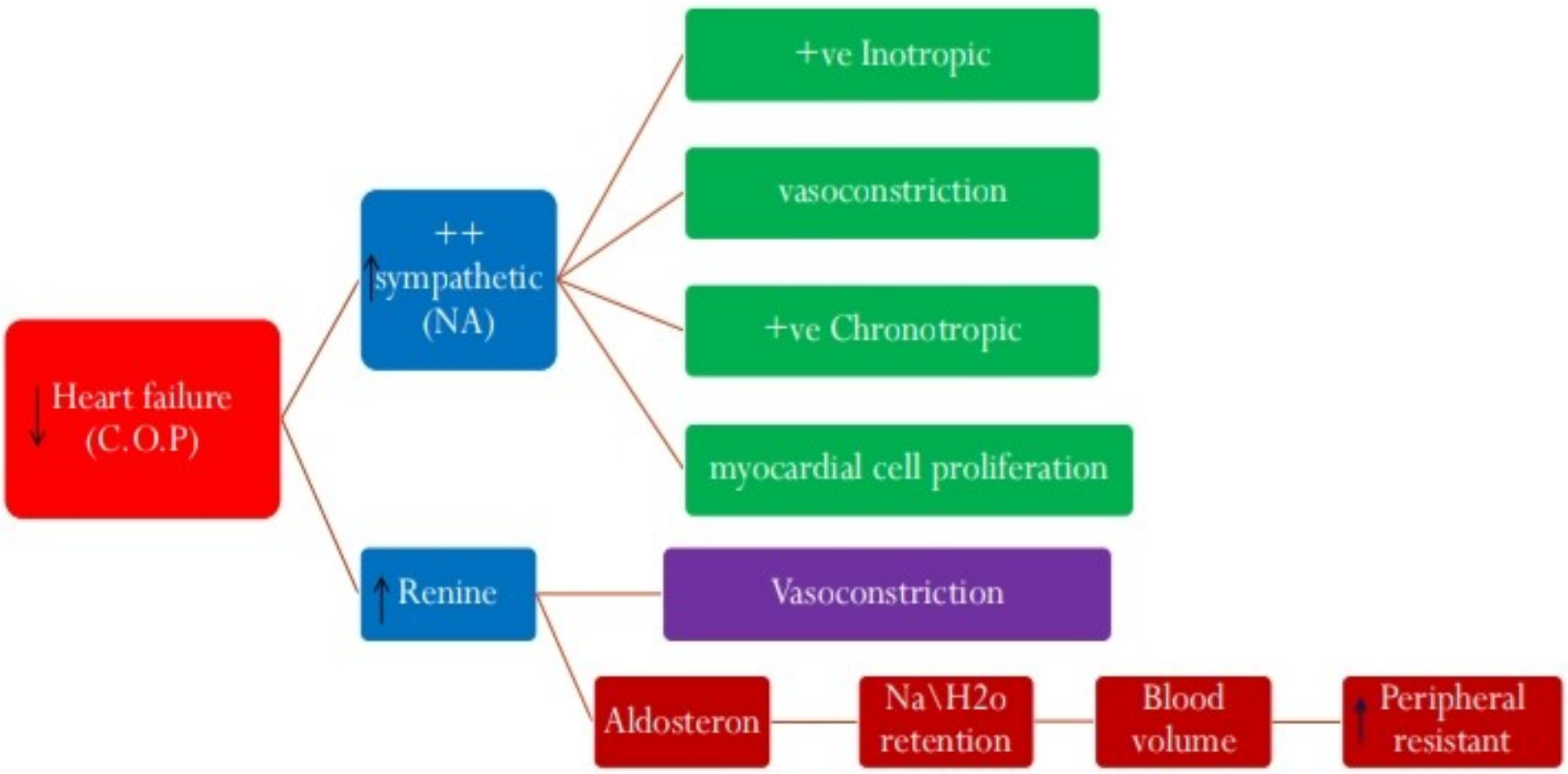
➢ Pulmonary congestion
➢ Dyspnea
➢ Fluid retention
➢ Fatigue

Causes of Heart Failure

➢ Dysfunction of contractile myocardial cells and the endothelial cells that line the heart and blood vessels.

➢ Hyperthyroidism
➢ Excessive intravenous fluids.
➢ Blood transfusions.
➢ Drugs that decrease the force of myocardial contraction (eg, antidysrhythmic drugs)
➢ Drugs that cause retention of sodium and water (eg, corticosteroids, estrogens, nonsteroidal anti- inflammatory agents).

Compensatory Mechanisms



Aims of treatment

➢ Alleviate symptom
➢ Improve survival

Therapeutic strategies in CHF

➢ Cardiac contractility (the principal therapeutic effect) (cardiac glycosides)

➢ ACE inhibitors
➢ Diuretics

➢ Additional Agents involve more aggressive diuretic therapy, vasodilators, parenteral inotropic agents (e.g. dobutamine).

➢ Beta adrenergic blocking agent

A- Cardiac glycosides (Digitalis)

E.g. **digoxin and digitoxin**

- **Mechanism of inotropic effect**

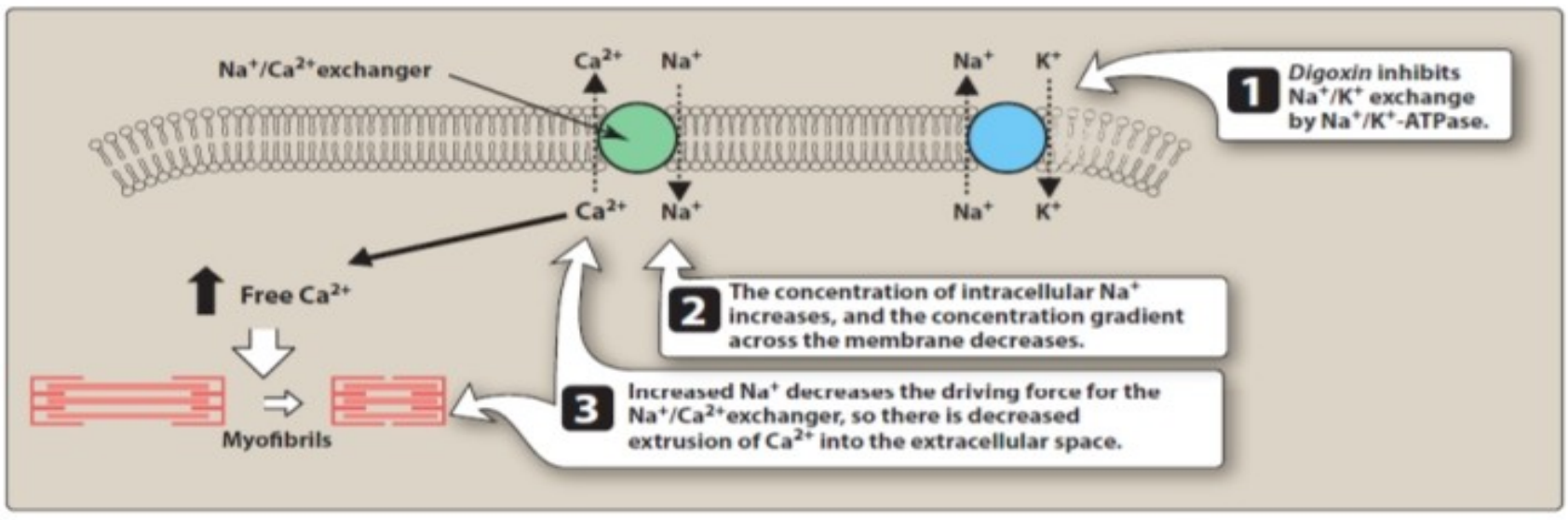
Na-K-ATPase pump inhibition

↓

↑ intracellular sodium3

↓

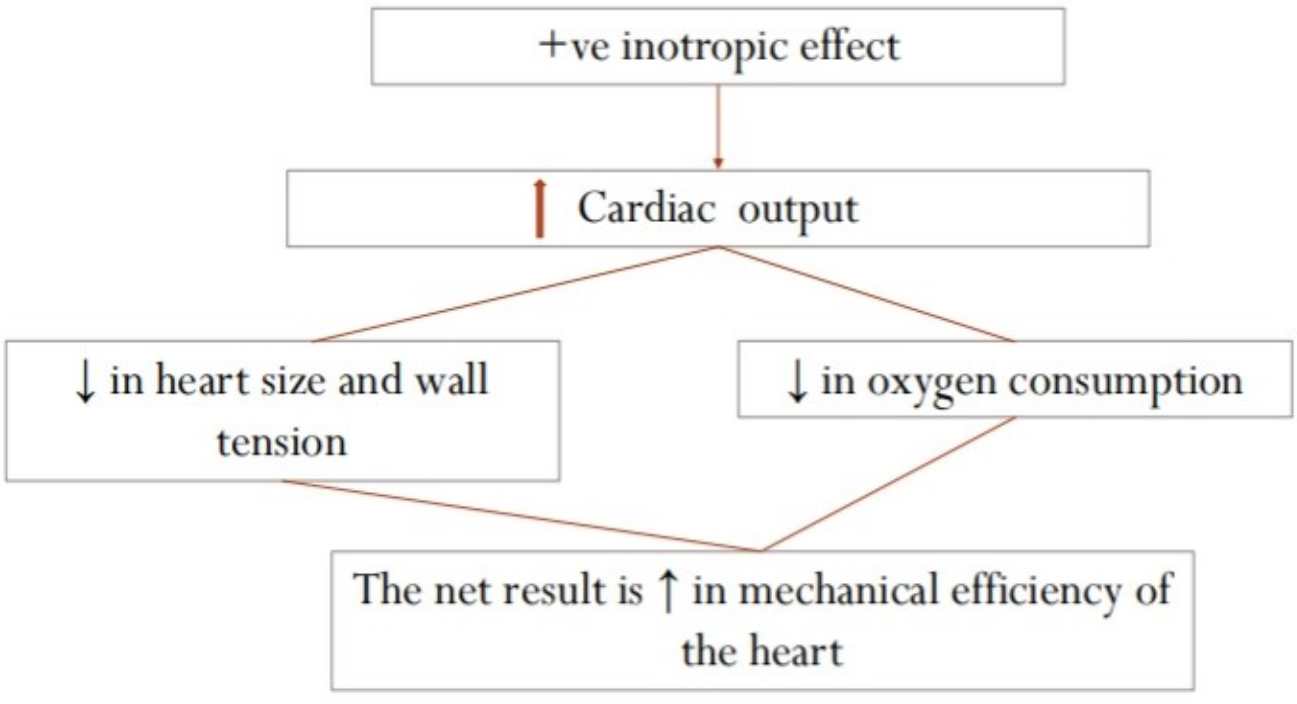
↑ calcium concentrations (+ve) inotropic responses.



- **Pharmacokinetics**

Glycoside	Oral bioavailability	Lipid solubility	Metabolism & excretion	Half-life	Time to peak
Digoxin	75%	Medium	Metabolized in liver (Hepatic) excreted in bile duct (Biliary)	36 h (1.5 days)	3-6h
Digitoxin	90-100%	High		5-7 days	6-12h

- **Pharmacological effects**



Pharmacological effects

➢ -ve chronotropic either directly or through vagal stimulation

-ve dromotropic

↓

slow conduction velocity and ↑ refractory period in atrium and AV node

Pharmacological effects

It deactivates renin-angiotensin aldosterone

↓

diuresis leading to ↓ fluid volume and edema

↓

↓ fluid retention

- **Therapeutic uses**

➢ Severe HFrEF after initiation of ACE inhibitor, β-blocker, and diuretic therapy.

- **Adverse effects**

1. ➢ **it has a very narrow therapeutic index, and digoxin toxicity**
    i. ➢ ***DIGOXIN 0.25 mg tab., 0.05 mg/ml pediatric elixir, 0.5 mg/2 ml inj.***
2. ➢ **Anorexia**
3. ➢ **nausea**
4. ➢ **vomiting**
5. ➢ **Blurred vision**
6. ➢ **yellowish vision (xanthopsia)**
7. ➢ **Hypokalemia**
8. ➢ **Cardiac arrhythmias**
9. ➢ **Ventricular tachycardia**

- **Precautions and interactions**

➢ Potassium antagonizes digitalis effects

➢ Hypokalemia (↑ digoxin binding to cells (↑ toxicity so K-sparing diuretic or K-supplement should take with digitalis.

➢ clarithromycin, verapamil, and amiodarone, can significantly increase digoxin levels, necessitating a reduced dose of digoxin.

➢ Digoxin should also be used with caution with other drugs that slow AV conduction, such as β-blockers, verapamil, and diltiazem.

- **Amrinone and Milrinone**

➢ The two most important actions of amrinone are positive inotropy and direct vasodilatation.

➢ Inhibit phosphodiesterase enzyme

↓

➢ Increases myocardial cAMP and transmembrane influx of Ca2+

↓

➢ ↑ myocardial contractility

➢ Amrinone is indicated only for short-term i.v. use in severe and refractory CHF,
➢ Milrinone is the drug of choice for short term parenteral

- **Adverse effects**

➢ Thrombocytopenia

Sympathomimetic inotropic drugs

1. ➢ Dobutamine
2. ➢ Dopamine

Dobutamine

➢ a relatively selective β1 agonist with prominent inotropic action

➢ is the preferred drug for i.v. infusion in acute heart failure accompanying myocardial infarction (MI)

Dopamine

➢ used in cardiogenic shock due to MI and other causes.
➢ Induce vasodilation by direct stimulation of D1 and D2 post-synaptic dopamine receptors.
➢ Vasodilatation of renal vasculature (↑ GFR (may benefit patients with poor renal perfusion)

Angiotensin Converting Enzyme Inhibitors

➢ Drug of choice in the treatment of hypertensive patients with CHF.

Beta-Adrenergic Blocking Agents

1. Bisoprolol
2. Carvedilol
3. Metoprolol

Suppress activation of the sympathetic nervous system
↓
Heart returns toward a more normal shape and function
↓
Cardiac output increases

➢ Started once normal blood volume is restored and edema and other symptoms are relieved.

➢ Added to the ACE inhibitor/diuretic regimen, usually near the end of a hospital stay or as outpatient therapy.

➢ Not recommended for clients in acute HF because of the potential for an initial decrease in myocardial contractility.

Human Natriuretic Peptide B-type

➢ B-type natriuretic peptide, which is secreted primarily by the ventricles in response to fluid and pressure overload.
➢ Increases diuresis and secretion of sodium
➢ Suppressing the renin–angiotensin–aldosterone system.
➢ Decreasing secretion of the neurohormones endothelin and norepinephrine.

Nesiritide

➢ Nesiritide is identical to endogenous human B-type natriuretic peptide
➢ Is the first in this new class of drugs to be used in the management of acute HF. Produced by recombinant DNA technology.

Endothelin Receptor Antagonists

Bosentan

- This new class of drugs relaxes blood vessels and improves blood flow by targeting endothelin-1 (a neurohormone) that is produced in excess in heart failure

Hypo-lipidaemic Drugs

Hypo-lipidaemic Drugs

➢Prevent cardiovascular disease by retarding the accelerated atherosclerosis in hyperlipidaemic individuals.

Hyperlipidemia (Hyperlipoproteinaemia)

➢Lipids and cholesterol are transported through the bloodstream as macromolecular complexes of lipid and protein known as **lipoproteins**.

➢There are four main classes of lipoprotein;-

1. ➢Chylomicrons
2. ➢High density lipoproteins (HDL)
3. ➢Low density lipoproteins (LDL)
4. ➢Very low density lipoproteins (VLDL)

Characteristics and function of plasma lipoproteins

Lipoprotein	Lipid contained	Source of lipid	Function
Chylomicron	TG >> CHE	Diet	Dietary TG transport
Chylomicron remnant	CHE >> TG	Diet,Chy	Dietary CH transport
VLDL	TG >> CHE	Liver	EndogenousTG transport
LDL	CHE > TG	VLDL	Transport CHE & TG to liver, source of LDL
	CHE	LDL	Transport CH & to Tissues and liver
HDL	Phospholipid, CHE	Tissues, cell memb.	Removal of CH from tissue

Types of primary hyperlipoproteinaemias

Type Disorder	Cause	Elevated plasma lipoprotein	Plasma lipids	
			CH	TG
I-Familial lipoprotein lipase deficiency	Genetic	Chylomicron	↑↑	↑↑↑
IIa Familial hypercholesterolaemia	Genetic	LDL	↑↑	N
IIb Polygenic hypercholesterolaemia	Multifactorial	LDL	↑	N
III Familial dysbetalipoproteinaemia	Genetic	IDL, Chy. rem.	↑	↑
IV Hypertriglyceridaemia	Multifactorial, Genetic	VLDL	N	↑↑
V Familial combined hyperlipidaemia	Genetic	VLDL, LDL	↑	↑

Causes of hyperlipidemia

1. Primary

- Genetic defect
- Dietary and physical activity

Secondary

- Diabetes
- Myxoedema
- Nephrotic syndrome
- Chronic alcoholism
- Drugs (corticosteroids, oral contraceptives, β-blockers)

Management of Hyperlipidemia

1) Diet
2) Lifestyle changes
3) Midication

Diet

- Restriction of caloric intake (35% as fat, 50% as carbohydrate and 15% as protein)
- Restriction of cholesterol to < 300 mg/day (equivalent to 1½ eggs)
- Replace animal fats with polyunsaturated fatty acids
- ↑ dietary fibers (fruits & vegetables)

Lifestyle changes

1) Weight reduction
2) Exercise
3) Cessation of smoking
4) Reduction or elimination of alcohol intake

Classification Of Hypo-lipidemic Drugs

1. HMG-CoA reductase inhibitors (Statins)
2. Bile acid sequestrants (Resins)
3. Lipoprotein lipase activators (PPARα activators, Fibrates)
4. Lipolysis and triglyceride synthesis inhibitor
5. Sterol absorption inhibitor

1. HMG-CoA reductase inhibitors (Statins)

- They competitively inhibit conversion of 3-Hydroxy-3-methyl glutaryl coenzyme A (HMG-CoA) to mevalonate by the enzyme HMG-CoA reductase.

Decreased cholesterol synthesis
↓
Increased receptor mediated uptake and catabolism of IDL and LDL
↓
Compensatory increase in LDL receptor expression on liver cells

- Therapeutic doses reduce CH synthesis by 20–50%.

1. HMG-CoA reductase inhibitors (Statins)

- Lovastatin
- Simvastatin
- Pravastatin
- Atorvastatin
- Rosuvastatin
- Pitavastatin

Effect on lipids (%)

1. ↓LDL (20–55)
2. ↑HDL (5–15)
3. ↓TG (10–35)

➢ Because HMG-CoA reductase activity is maximum at midnight, all statins are administered at bed time to obtain maximum effectiveness.

Lovastatin

➢ It is the first clinically used statin.
➢ First pass metabolism is extensive,
Metabolites are excreted mainly in bile.

➢ Dose: 10–40 mg/day (max. 80 mg).

Simvastatin

➢ It is twice as potent as lovastatin; also more efficacious.
➢ A greater rise in HDL-CH (when low) has been noted with simvastatin than lovastatin or pravastatin

➢ Oral absorption is better.
➢ First pass metabolism extensive, t½ is 2–3 hr.
➢ Dose: 5–20 mg/day (max. 80 mg)

Atorvastatin

➢The Newer and most popular statin
➢ is more potent and appears to have the highest LDL-CH lowering efficacy at maximal daily dose of 80 mg.
➢ a greater reduction in TGs is noted
➢ Atorvastatin has a much longer plasma t½ of 18–24 hr than other statins, and has additional antioxidant property.
➢ Dose: 10–40 mg/day (max. 80 mg)

Rosuvastatin

➢ This is another newer, commonly used and potent statin (10 mg rosuvastatin ~ 20 mg atorvastatin)
➢ a plasma t½ of 18–24 hours.

➢ Greater LDL-CH reduction can be obtained in severe hypercholesterolaemia
➢ In patients with raised TG levels, rosuvastatin raises HDL-CH by 15–20% (greater rise than other statins).
➢ Dose: Start with 5 mg OD, increase if needed upto 20 mg/day, (max 40 mg/day)

Pitavastatin

➢ This is the latest and dose-to-dose the most potent statin.
➢ A ceiling response of 40% LDL-CH reduction with the maximum recommended daily dose of 4 mg is noted.
➢ The plasma t½ is 12 hours.
➢ Dose: 1–4 mg per day;

☒ Therapeutic uses

➢ The first choice drugs for primary hyperlipidaemias with raised LDL and total CH levels, with or without raised TG levels (Type IIa, IIb, V)
➢ Secondary (diabetes, nephrotic syndrome) hypercholesterolemia.

☒ Adverse effects

➢ Transient elevation of serum transaminases.
Monitoring of liver function is recommended.

➢ Myopathy (muscle weakness) and rhabdomyolysis (disintegration of muscle tissue).

☒ Contraindications

1. ➢ Pregnancy and lactation
2. ➢ Children and teenagers
3. ➢ Homozygous familial hypercholesterolemia

2. Bile acid sequestrants (Resins)

1. Cholestyramine
2. Colestipol

2. Bile acid sequestrants (Resins)

bind bile acids in the intestine
↓
interrupting their enterohepatic circulation
↓
Faecal excretion of bile salts is increased.

 leads to

➢ leads to enhanced hepatic metabolism of CH to bile acids
➢ More LDL receptors are expressed on liver cells:
➢ Clearance of plasma IDL, LDL.

 Therapeutic uses

➢ Primary chylomicronemia
➢ Familial hypertriglyceridemia
➢ Relief of excess bile acid-induced pruritus in patients with biliary obstruction

 Adverse effects

➢ Cause flatulence and other g.i. symptoms
➢ Impairs the absorption of fat-soluble vitamins (ADEK).
➢ Interfere with absorption of many drugs and have poor patient acceptability

3. Lipoprotein lipase activators
peroxisome proliferator-activated receptor-α
(PPARα activators, Fibrates)

1. ➢ Gemfibrozil 2. ➢ Bezafibrate 3. ➢ Fenofibrate

➢ These drugs activate the nuclear peroxisome proliferator-activated receptor-α (PPAR α) which is found mainly in hepatocytes and adipocytes and functions to regulate lipoprotein metabolism.

Activation of PPAR α lead to :-

➢ Enhances lipoprotein lipase synthesis and fatty acid oxidation.
➢ Enhanced LDL receptor expression in liver
➢ Decrease hepatic TG synthesis

➢ lower TG levels by 20–50%,
➢15%–10 decrease in LDL-CH
➢ a 10–15% increase in HDL-CH.

 Therapeutic uses

➢ Indicated in clinical disorders associated with ↑ circulating TGs
➢ suitable choice for the treatment of dyslipidemia in diabetic patients who often demonstrate hypertriglyceridemia and low levels of HDL.

 Adverse effects

➢ *GIT disturbances* 10%–5) of patients)
➢ *Cholelithiasis (gallstone formation)* due to ↑ biliary cholesterol excretion.
➢ *Myopathy*

- **Contraindications**

1. ➢ Pregnancy and lactation
2. ➢ Severe hepatic and renal dysfunction
3. ➢ Pre-existing gallbladder disease

4. Lipolysis and triglyceride synthesis inhibitor

Nicotinic Acid (Niacin)
Vitamin B3

reduces production of VLDL in liver by inhibiting TG synthesis.

↓

VLDL degradation products IDL and LDL are also reduced.

- **Therapeutic uses**

➢ Familial hyperlipidemias (↓ cholesterol & TGs).
➢ Severe hypercholsterolemias (+ statins).
➢ Familial dysbetalipoproteinemia (Type III)
➢ Its most important indication is to control pancreatitis associated with severe hypertriglyceridaemia,

- **Adverse effects**

➢ Flushing (due to cutaneous PG-mediated vasodilatation)
➢ Dyspepsia, nausea & abdominal pain.
➢ Uric acid retention

5. Sterol absorption inhibitor

Ezetimibe

➢ It is a novel drug that acts by inhibiting intestinal absorption of cholesterol and phytosterols.

↓ absorption of cholesterol
↓
↓ ch in chylomicrons
↓
↓ cholesterol delivery to the liver
↓
↑ hepatic LDL receptors
↓
↑ cholesterol clearance from blood.

- **Therapeutic uses**

➢ Treatment of hypercholesterolemia often in combination with a statin